

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Environmental Chemistry - Core Application	96 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Environmental Chemistry - Core Application (Very High Yield — 96 PYQs)

Core Principle: Key problem pattern in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "The possibility of photochemical smog formation will be minimum at :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Which of the following compounds is an example of Freon?..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Which of these reactions is not a part of breakdown of ozone in stratosphere?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

The possibility of photochemical smog formation will be minimum at :

- (A) Mumbai in May
- (B) Kolkata in October
- (C) New-Delhi in August (Summer)
- (D) Srinagar, Jammu and Kashmir in January

Correct Answer: None

Solution Analysis: The formation of photochemical smog depends on several factors, including sunlight, temperature, and the presence of pollutants such as nitrogen oxides and volatile organic compounds. Option (A) Mumbai in May may also have some possibility of photochemical smog formation because the temperature and sunlight are relatively high in May. Option (B) Kolkata in October may have some possibility of photochemical smog formation because the temperature and sunlight are still relatively high in October. Option (C) New-Delhi in August (Summer) is more likely to have photochemical smog formation because